James E. Hanson

Contact Information

Iowa State University
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Employment

Iowa State University

Assistant Professor, Fall 2024 to present

University of Maryland, College Park

Novikov Postdoctoral Fellow, Fall 2021 to Spring 2024

I was offered a postdoctoral position at the ANR Project AGRUME for Spring 2021 working with Tomás Ibarlucía at the Université de Paris, but the pandemic prevented me from officially accepting the position. Instead, I worked remotely with Ibarlucía on a related research project.

Education

University of Wisconsin–Madison

PhD in Mathematics, December 2020
MA in Physics, December 2016

University of Minnesota, Twin Cities

BSc in Mathematics and Physics, May 2012

Teaching

Iowa State University

- MATH 5010-01: Introduction to Real Analysis. Fall 2026.
- MATH 5900-03: Independent Study. Fall 2026 (Independent Study).
- MATH 6030-01: Mathematical Logic II. Spring 2026.
- MATH 1650-02: Calculus I. Fall 2025.
- MATH 6010-01: Mathematical Logic. Fall 2025.
- MATH 1660-5: Calculus II. Spring 2025.
- MATH 5010: Introduction to Real Analysis. Fall 2024.

Research Interests

My research interests lie primarily in model theory and in the intersection of logic and real analysis. In the domain of model theory, I am interested in neostability theory and continuous logic. Neostability theory attempts to broaden the applicability of model-theoretic ideas to broader classes of mathematical structures. Continuous logic has applications in analysis, such as ergodic theory and $C^\star$- and other operator algebras, and in some parts of algebra, such as Berkovich analytic geometry. I am also interested in computable analysis and computable structure theory.

Grants

National Science Foundation, Individual Research Grant, Principal Investigator, "Neostability and Continuous Logic" ($150,000 for three years, August 2026–July 2029).

Publications

1. J. E. Hanson, *Any function I can actually write down is measurable, right?* Expositiones Mathematicae, August 2025. `10.1016/j.exmath.2025.125718`.
2. J. Hanson, *Strongly Minimal Sets and Categoricity in Continuous Logic.* Memoirs of the American Mathematical Society, June 2025. `10.1090/memo/1574`.
3. J. E. Hanson, *Topometric characterization of type spaces in continuous logic.* Journal of Logic and Analysis, February 2025. `10.4115/jla.2025.17.1`.
4. J. E. Hanson, *Approximate categoricity in continuous logic.* Archive for Mathematical Logic, December 2024. `10.1007/s00153-024-00952-3`.
5. J. E. Hanson, *A simple continuous theory.* Journal of Mathematical Logic, October 2024. `10.1142/S0219061324500260`.
6. J. E. Hanson, *Bounded ultraimaginary independence and its total Morley sequences.* Model Theory, April 2024. `10.2140/mt.2024.3.39`.
7. J. E. Hanson, *Some semilattices of definable sets in continuous logic.* Journal of Logic and Analysis, April 2024. `10.4115/jla.2024.16.3`.
8. J. E. Hanson, *Approximate isomorphism of metric structures.* Mathematical Logic Quarterly, September 2023. `10.1002/malq.202200076`.
9. G. Conant, K. Gannon, J. Hanson, *Keisler measures in the wild.* Model Theory 2, no. 1 (2023): 1-67. `10.2140/mt.2023.2.1`.
10. J. Hanson, *Metric spaces are universal for bi-interpretation with metric structures.* Ann. Pure Appl. Logic (2023). `10.1016/j.apal.2022.103204`.
11. J. Hanson, T. Ibarlucía, *Approximate isomorphism of randomization pairs.* Confluentes Mathematici, Volume 14 (2022) no. 2, pp. 29-44. `10.5802/cml.85`.
12. G. Conant, J. Hanson, *Separation for isometric group actions and hyperimaginary independence.* Fundamenta Mathematicae 259 (2022), 97-109. `10.4064/fm167-2-2022`.
13. J. Hanson, *Analog reducibility.* Journal of Logic and Computation, 2021, exab036. `10.1093/logcom/exab036`.
14. W. Cottrell, J. Hanson, A. Hashimoto, A. Loveridge, and D. Pettengill, *Intersecting D3-D3′ brane system at finite temperature.* Phys. Rev. D, 95, 044022 (2017). `10.1103/PhysRevD.95.044022`.
15. W. Cottrell, J. Hanson, and A. Hashimoto, *Dynamics of* $\mathcal{N} = 4$ *supersymmetric field theories in 2 + 1 dimensions and their gravity dual.* J. High Energ. Phys. 2016, 12. `10.1007/JHEP07(2016)012`.

Publications (Preprints)

1. J. E. Hanson, *A natural haystack of differentially closed fields.* `arxiv.org/abs/2606.28663`.
2. J. Arulseelan and J. E. Hanson, *Pointwise Mean Value Theorems in Constructive Mathematics.* `arxiv.org/abs/2606.21738`.
3. H. Jeon and J. E. Hanson, *The Axiom of Double Complement and its opposites.* `arxiv.org/abs/2606.00861`.
4. J. E. Hanson and C. Watson, *A formula for any real number, maybe.* `arxiv.org/abs/2602.02384`.
5. J. E. Hanson, T. Koberda, J. de la Nuez González, and C. Rosendal, *Set theory, logic, and homeomorphism groups of manifolds.* `arxiv.org/abs/2512.05206`.
6. N. Ackerman, C. Freer, K. Gannon, J. E. Hanson, and R. Patel, *Generic sampling and invariant measures on the space of k-uniform hypergraphs.* `arxiv.org/abs/2510.17090`.

7. B. Bodor, S. Braunfeld, and J. E. Hanson, *Labelled growth rates of ω-categorical structures and applications in choiceless set theory*. arxiv.org/abs/2509.12656.
8. J. E. Hanson, *A combinatorial characterization of Kim's lemma for pairs of bi-invariant types*. arxiv.org/abs/2507.21366.
9. J. E. Hanson, *Indiscernible extraction at small large cardinals from a higher-arity stability notion*. arxiv.org/abs/2506.19147. Submitted.
10. A. Bauer, J. E. Hanson, *The countable reals*. arxiv.org/abs/2404.01256. Submitted.
11. J. E. Hanson, *Uniqueness of constructible models in continuous logic*. arxiv.org/abs/2501.02679.
12. K. Gannon, J. E. Hanson, *Model theoretic events*. arxiv.org/abs/2402.15709. Submitted.
13. G. Conant, K. Gannon, J. E. Hanson, *Generic stability, randomizations, and NIP formulas*. arxiv.org/abs/2308.01801. Submitted.
14. J. E. Hanson, *Bi-invariant types, reliably invariant types, and the comb tree property*. arxiv.org/abs/2306.08239. Submitted.
15. J. Hanson, *A metric set theory with a universal set*. arxiv.org/abs/2302.02258. Submitted.

Talks

Small large cardinals and neostability theory
- New Forking Festival at the University of Notre Dame. (July 2026)
- 2026 North American Annual Meeting of the Association for Symbolic Logic, University of Pennsylvania Carey Law School. (July 2026)
- Logic Colloquium 2026 at Swansea University. (July 2026)
- BLAST 2026, Baylor University. (May 2026)

Special coheirs and model-theoretic trees (or sometimes *Forcing with model-theoretic trees*)
- UW-Madison Logic Seminar. (March 2026)
- Logic Colloquium 2025 at TU Wien. (July 2025)
- ICICL Summer Research Conference at Drake University. (June 2025)
- UIC Logic Seminar. (April 2025)
- PKU Model Theory Seminar. (November 2024)
- Iowa State University Logic Seminar. (September 2024)
- 2024 AMS Spring Central Sectional Meeting Special Session on Model Theory, University of Wisconsin–Milwaukee. (April 2024)
- University of Maryland Logic Seminar. (October 2023)

Measurability properties of model-theoretic invariant types
- Iowa State University Logic Seminar. (September 2025)

How bad could it be? The semilattice of definable sets in continuous logic
- Iowa State University Logic Seminar. (March 2025)
- NYLogic Logic Workshop. (April 2023)
- Wesleyan Logic Colloquium. (November 2022)
- University of Maryland Logic Seminar. (October 2022)

Independence in arbitrary theories via automorphism groups and large cardinals
- SEALS 2024, University of Florida. (March 2024)

Tameness and definability in continuous logic
- Iowa State University. (January 2024)

Generic stability and randomizations
- Mid-Atlantic Mathematical Logic Seminar Spring Fling 2023. (May 2023)
- Joint Mathematics Meetings. (April 2022)

Bounded ultraimaginary independence
- 2023 North American Annual Meeting of the ASL. (March 2023)
- UCLA Logic Colloquium. (May 2022)
- University of Maryland Logic Seminar. (February 2022)

An introduction to continuous logic
- VCU Analysis, Logic and Physics Seminar. (April 2022)

Definable sets in continuous logic
- University of Maryland Logic Seminar. (September 2021)

A gentle introduction to continuous logic
- University of Maryland Logic Seminar. (September 2021)

A Versatile Counterexample for Invariant Types and Keisler Measures outside NIP
- Notre Dame Model Theory Seminar (Digital). (March 2021)
- Séminaire de Logique Lyon-Paris (Digital). (March 2021)

Joint work with Gabriel Conant and Kyle Gannon.

Strongly Minimal Sets in Continuous Logic (regarding essential continuity)
- Online Logic Seminar (Digital). (August 2021)

Definability and Categoricity in Continuous Logic
- UW Logic Seminar (Digital), University of Wisconsin–Madison. (April 2020)

Skolemization in Continuous Logic
- UW Logic Seminar, University of Wisconsin–Madison. (November 2019)

Strongly Minimal Sets in Continuous Logic (regarding categoricity)
- Logic Seminar, University of Illinois at Chicago. (October 2019)
- AMS Sectional Meeting, Special Session on Model Theory, University of Wisconsin–Madison. (September 2019)
- Graduate Student Conference in Logic XX, University of Illinois at Chicago. (April 2019)
- UCI Logic and Set Theory Seminar, University of California, Irvine. (April 2019)
- UW Logic Seminar, University of Wisconsin–Madison. (February 2019)

Separable and inseparable Gromov-Hausdorff categoricity in continuous logic
- Association for Symbolic Logic North American Annual Meeting, Western Illinois University. (May 2018)

- Graduate Student Conference in Logic XIX, University of Wisconsin–Madison. (April 2018)

Encoding metric structures as metric spaces
- UW Logic Seminar, University of Wisconsin–Madison. (February 2018)

Workshops

A Convergence of Computable Structure Theory, Analysis, and Randomness, March 19-March 24, 2023, in Banff.

Service

Institutional Service
- Co-organizer, Iowa State Mathematics Research Teams (ISMaRT). (Fall 2025–present)
- Co-organizer, ISU Logic Seminar. (Fall 2024–present)
- Member, Iowa Colloquium on Information, Complexity, and Logic (ICICL) Organization Committee. (2024–present)
- Opt-In Committee, Department of Mathematics, Iowa State University. (2025–present)
- Volunteer, Iowa State Mathematics Department Sonia Kovalevsky Day. (April 2025)
- Reviewer, GATEway Science Fair, Rochester, Minnesota. (April 2025, April 2026)
- Reviewer, State Science & Technology Fair of Iowa. (March 2026)

Professional Service
- Organizer, AMS Special Session on Model Theory, 2026 Fall Central Sectional Meeting. (October 2026)
- Referee for *Advances in Mathematics*, *Applied Categorical Structures*, *Journal of Symbolic Logic*, and *Notre Dame Journal of Formal Logic*. (2022–2025)
- External member, master's thesis committee for Connor Watson, Florida Atlantic University. (2025)

Awards

University of Wisconsin-Madison, Department of Mathematics
- *Excellence in Mathematical Research Award*, October 2019.
- *Physical Sciences Award*, October 2018.

University of Wisconsin-Madison, Department of Physics
- *Van Vleck Fellowship for Teaching Assistants*, September 2012.
- *David L. Huber Fellowship*, September 2012.
- *Firminhac Fellowship*, September 2012.

University of Minnesota
- *Professor Hans H. Dalaker Scholarship for Undergraduate Mathematics*, April 2011.
- *Presidential Scholarship*, August 2008.
- *Maroon and Gold Leadership Award*, August 2008.
- *National Merit Scholarship*, August 2008.
- *Undergraduate Research Scholarship*, August 2008.